Sebastian Ren

UCL Robotics & AI MEng

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EDUCATION

University College London Sep 2023 – Jun 2027

MEng Robotics and Artificial Intelligence, Department of Computer Science London, UK

- **Average:** 74.1% (First Class)

PUBLICATIONS

[1] **Ren, Y.**†, Toh, B.-Z.†, Yu, T. A., Ong, N. K., Mao, Z., & Bano, S. "CasCVS-Net: A Staged Multi-Task Cascade for Critical View of Safety Assessment." (†Equal contribution) **MICCAI SafeSurg 2026 (Oral).** [Paper]

[2] Pan, Y., **Ren, Y.**, Chan, C., Sun, J., & Zhou, C. "Calf-Integrated Arms for Bimanual Quadruped Loco-Manipulation." **CLAWAR 2026 (Oral).**

[3] Xu, Y., Zhu, F., Li, Y., **Ren, S.**, Huang, X., & Chen, Y. "RGBSQGrasp: Inferring Local Superquadric Primitives from Single RGB Image for Graspability-Aware Bin Picking." **IROS 2025 RGMCW (Best Workshop Paper Award).**

RESEARCH & INTERNSHIP EXPERIENCE

Robotics Engineer – BYTE ABEE Jun – Sep 2026

Humanoid Locomotion & Motion Control Athens, Greece

- Fine-tuned SONIC to transfer humanoid motion-control capabilities from **Unitree G1 to AgiBot X2**.
- Deployed the adapted controller on the **physical X2 robot** for motion execution.
- Built a camera-based motion-imitation system enabling X2 to observe and reproduce a human demonstrator's movements.

Research Project – UCL Hawkes Institute Apr – Jun 2026

CasCVS-Net: Surgical Scene Understanding & Safety Assessment London, UK

- Co-authored CasCVS-Net (equal contribution), coupling detection, segmentation, and surgical Critical View of Safety (CVS) assessment through predicted anatomy and staged multi-task training.
- Achieved **67.2 CVS mAP** on Endoscapes (**+4.0 points** over LG-CVS); improved rare-anatomy segmentation from 4.6 to 15.3 mIoU over a matched single-task baseline.

Research Project – UCL Robot Perception & Learning Lab Nov 2025 – Jan 2026

Bipedal Locomotion for Quadrupedal Robots London, UK

- Reproduced TumblerNet for bipedal locomotion of quadrupedal robots in Isaac Gym simulation, using PyTorch and PPO.
- Implemented the paper's heterogeneous actor–critic, RMA-style state estimator, and reward functions based on VHIP and Cart-Table dynamics.

Research Intern – ANTOBOT Mar – May 2025

Human-Following Mobile Robot System London, UK

- Built a ROS person-following system with YOLOv8n and OSNet Re-ID; fused vision and odometry with an EKF to maintain target state estimates through occlusions.
- Integrated an Artificial Potential Field controller for obstacle avoidance; validated in Gazebo and deployed on a physical Antobot robot.

Research Intern – The Future Laboratory, Tsinghua University Jun – Sep 2024

Olfaction & Bio-Robotics Beijing, China

- Simulated airflow around insect antennae in Ansys Fluent to study bio-inspired airflow sensing.
- Developed E-nose sensor data collection software and led experimental setup design.

AWARDS & HONORS

- **Kaggle 18th Place (Silver Medal)**, Deep Past Challenge: Akkadian–English Translation 2026
- **Gold Medal**, UKMT Mathematical Challenge 2023
- **Gold Medal**, British Physics Olympiad 2021